Sagnac effect

Preparing for the first meeting

Read a bit about the following topics and system components:

1. Polarizers
2. Lasers
3. Beam splitter
4. Wave interference
5. Gaussian beam
6. Single mode fiber
7. Berry phase → Examples of geometric phases → Polarized light in an optical fiber

In addition, read the Sagnac manual from moodle and the following paper - Eyal Schwartz, 2017 Eur. J. Phys. **38** 015301 (Sagnac effect in an off-center rotating ring frame of reference)

Your pre-report is composed of answers to the following questions:

1. Derive an expression for the interference of two plane wave beams that arrives at a screen with a relative angle.

2. From the previous derivation, express the distance between adjacent maxima using the wavelength of the light and the angle between the beams.

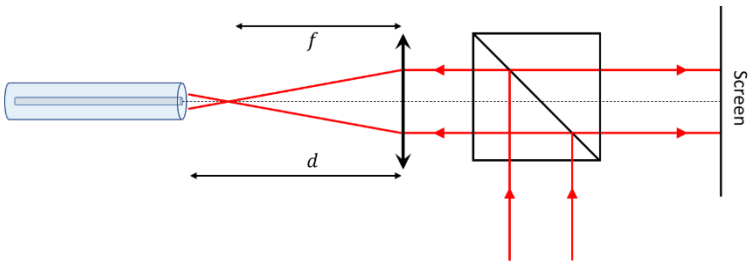
3. Write down the expression for a gaussian beam.

4. Derive an expression for the Sagnac formula: $\Delta\phi(\omega) = N \cdot \frac{8\pi}{\lambda c} A \cdot \omega$.

5. Explain how you can extract $\Delta\phi(\omega)$ from the measured interference pattern (write two different ways to do the analysis). Assuming that $\Delta\phi_{max}$ is large, which way is more accurate?

6. Explain how to reduce unwanted reflections using two polarizers (Malus law) and how to increase the wanted signal using Berry phase.

7. Use geometrical optics to explain the influence of changing the distance (d) between the lens and the fiber for light coming out of the fiber and light that is reflected from the fiber surface. Assume that light comes out from the fiber as a point source.



Schematic for reflected light from the fiber surface, the angles of the reflected light are not necessarily correct.

8. How will the center of rotation influence the Sagnac formula?

9. How the wavelength of light will influence the interference pattern?

10. How the refractive index influences the Sagnac formula (find the answer in - Eyal Schwartz, 2017 Eur. J. Phys. **38** 015301)

11. Write down the core diameter, Numerical aperture, and effective refractive index of refraction (n_{eff}) at wavelengths of 1550nm and 1310nm of the fiber you are going to use in the experiment (SMF-28e+).

12. Does SMF-28e+ is a single-mode fiber for 640nm light?